Model Evaluation & Comparison

Linear vs. Non-Linear Models for Heart Heart Disease Prediction

Course – Data Science & Big Data Analysis | B.Tech-CSE-AIML | Sec- ‘A’

Presented By:

- Prashant Sharma (SBU233821)
- Shubham Anand Pandey (SBU233747)
- Aryav Deep (SBU233709)
- Aditya Narayan Singh (SBU233758)
- Prabhat Kumar (SBU233738)
- Mansingh Mahto (SBU233384)

What is Model Evaluation?

Model evaluation is the systematic process of **assessing a machine learning model's performance on data it has never seen before**. It is one of the most critical steps in the data science pipeline — a model that performs well on training data but fails on new data is practically useless. Evaluation tells us whether a model has truly learned the underlying patterns in the data or has merely memorized the training examples.

The Core Challenge

During training, a model optimizes its parameters to minimize error on the training set. However, the real test is **generalization** — how well does the model perform on new, unseen data? A model that fits the training data too closely suffers from **overfitting**, while one that is too simple suffers from **underfitting**. Evaluation metrics help us detect and quantify both.

The Evaluation Workflow

- **Split the data** into training and test sets (or use cross-validation)
- **Train the model** on the training set only
- **Predict** on the held-out test set
- **Compute metrics** (Accuracy, Precision, Recall, F1, ROC-AUC)
- **Compare models** and select the best performer for deployment

This process ensures that our final model is not just theoretically sound, but practically reliable when deployed in real-world scenarios.

Why Do We Evaluate Models?

Model evaluation is not a formality — it is the foundation of trustworthy, deployable machine learning. In fields like healthcare, where incorrect predictions can have life-or-death consequences, rigorous evaluation is an ethical and professional obligation. Here is why it matters:

Build Trust in AI Systems

Stakeholders — doctors, patients, regulators — need to trust that an AI model's predictions are reliable. Quantitative evaluation metrics provide the evidence needed to build that trust and justify the model's use in clinical decision support systems.

Select the Best Algorithm

Not all algorithms are equally suited to every problem. By evaluating multiple models on the same dataset using the same metrics, we can objectively identify which approach — linear or non-linear — better captures the structure of the data.

Tune Hyperparameters

Evaluation metrics guide the hyperparameter tuning process. By measuring how changes in model configuration affect performance on a validation set, we can systematically optimize the model for the best possible generalization.

Ensure Safe Deployment

Before deploying any model in a production environment — especially in critical domains like healthcare — evaluation confirms that the model meets the required performance thresholds and will not cause harm due to systematic errors or biases.

Metrics Used for Evaluation

To comprehensively assess our models, we use five complementary metrics. Each metric captures a different aspect of model performance, and together they provide a complete picture of how well a classifier is working. No single metric tells the whole story.

Accuracy

Definition: The proportion of correct predictions (both true positives and true negatives) out of all predictions made.

Formula: (TP + TN) / (TP + TN + FP + FN)

Use Case: Best used when classes are balanced. Can be misleading in imbalanced datasets.

Precision

Definition: Of all instances predicted positive, what proportion were actually positive? Measures *exactness*.

Formula: TP / (TP + FP)

Use Case: Critical when **false positives are costly** — e.g., unnecessary medical procedures.

Recall (Sensitivity)

Definition: Of all actual positive instances, what proportion were correctly identified? Measures *completeness*.

Formula: TP / (TP + FN)

Use Case: Critical when **false negatives are costly** — e.g., missing a heart disease diagnosis.

F1-Score

Definition: The harmonic mean of Precision and Recall. Balances both metrics into a single score.

Formula: 2 × (Precision × Recall) / (Precision + Recall)

Use Case: Ideal when you need a **single balanced metric**, especially with class imbalance.

ROC-AUC Score

Definition: Area Under the Receiver Operating Characteristic Curve. Measures the model's ability to distinguish between positive and negative classes across all thresholds.

Range: 0.5 (random) to 1.0 (perfect)

Use Case: Excellent for **comparing models** independent of classification threshold.

Key Terms: TP = True Positive, TN = True Negative, FP = False Positive, FN = False Negative. In heart disease prediction, a **False Negative** (missing a diagnosis) is typically more dangerous than a False Positive.

The Heart Disease Dataset

Dataset Overview

The models in this study were trained and evaluated on a widely-used **clinical Heart Disease dataset** sourced from the UCI Machine Learning Repository. This dataset contains medical records of patients who underwent cardiac evaluation, with the primary goal of **binary classification**: predicting whether a patient has heart disease or not.

Target Variable: A binary label where **0 = No Heart Disease** and **1 = Heart Disease Present**.

Classification Task Type

This is a **supervised binary classification** problem. The model learns from labeled historical patient data and then predicts the class label for new, unseen patients.

Dataset Characteristics

- Binary target: 0 (No Disease) / 1 (Disease)
- Multiple clinical and demographic features
- Includes both continuous and categorical variables
- Standard benchmark for ML classification
- High practical relevance in healthcare

[4]:

	id	age	sex	dataset	cp	trestbps	chol	fbs	restecg	thalch	exang	oldpeak	slope	ca	thal	num
0	1	63	Male	Cleveland	typical angina	145.0	233.0	True	lv hypertrophy	150.0	False	2.3	downsloping	0.0	fixed defect	0
1	2	67	Male	Cleveland	asymptomatic	160.0	286.0	False	lv hypertrophy	108.0	True	1.5	flat	3.0	normal	2
2	3	67	Male	Cleveland	asymptomatic	120.0	229.0	False	lv hypertrophy	129.0	True	2.6	flat	2.0	reversable defect	1
3	4	37	Male	Cleveland	non-anginal	130.0	250.0	False	normal	187.0	False	3.5	downsloping	0.0	normal	0
4	5	41	Female	Cleveland	atypical angina	130.0	204.0	False	lv hypertrophy	172.0	False	1.4	upsloping	0.0	normal	0

Data Overview: Features & Preprocessing

Before training any model, the raw data must be carefully preprocessed. The quality of preprocessing directly impacts model performance. Below is a breakdown of the key features used and the transformations applied to make the data model-ready.

Key Input Features

Age

Patient's age in years — a continuous variable and strong risk factor

Sex

Binary categorical variable (Male / Female)

Chest Pain Type (cp)

Categorical: typical angina, atypical angina, non-anginal pain, asymptomatic

Resting BP (trestbps)

Resting blood pressure in mm Hg — continuous physiological measure

Cholesterol (chol)

Serum cholesterol in mg/dl — continuous biochemical marker

Preprocessing Pipeline

Raw clinical data rarely comes in a format directly usable by ML algorithms. The following preprocessing steps were applied to ensure both models received clean, standardized input:

1 One-Hot Encoding

Categorical variables such as **Chest Pain Type** and **Sex** were converted into binary indicator columns. This allows models to process non-numeric categories without imposing an artificial ordinal relationship.

2 Feature Scaling

Continuous features like **Age**, **Blood Pressure**, and **Cholesterol** were scaled to a standard range. Scaling is especially important for Logistic Regression, which is sensitive to the magnitude of input features.

3 Train-Test Split

The dataset was partitioned into a **training set** (used to fit the models) and a **held-out test set** (used exclusively for final evaluation), ensuring unbiased performance estimates.

Linear Model: Logistic Regression

What is Logistic Regression?

Despite its name, **Logistic Regression** is a classification algorithm, not a regression method. It is one of the most fundamental and interpretable models in machine learning, widely used as a **baseline classifier** for binary classification problems.

It models the **probability** that a given input belongs to a particular class using the logistic (sigmoid) function:

$$P(y = 1 \mid x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \ldots + \beta_n x_n)}}$$

The output is a probability between 0 and 1. A threshold (typically 0.5) is applied to convert this probability into a binary class prediction.

Why Use Logistic Regression as a Baseline?

→ **Simplicity & Interpretability**

Each coefficient directly indicates the direction and magnitude of a feature's influence on the outcome. This makes it highly interpretable — a critical property in medical applications where explainability matters.

→ **Linear Decision Boundary**

Logistic Regression assumes a **linear relationship** between features and the log-odds of the outcome. This is its primary limitation — it cannot capture complex, non-linear patterns in the data without manual feature engineering.

→ **Computationally Efficient**

It trains quickly even on large datasets and is less prone to overfitting on small datasets compared to more complex models. It serves as an excellent reference point for comparison.

Role in This Study: Logistic Regression serves as our **linear baseline model**.

Logistic Regression Performance

The Logistic Regression model was evaluated on the held-out test set using all five standard classification metrics. The results demonstrate **solid baseline performance**, with particularly strong Precision and ROC-AUC scores, indicating good discriminative ability despite the model's linear constraints.

80.43%

Accuracy

Overall correctness — 80.43% of test predictions were correct

86.14%

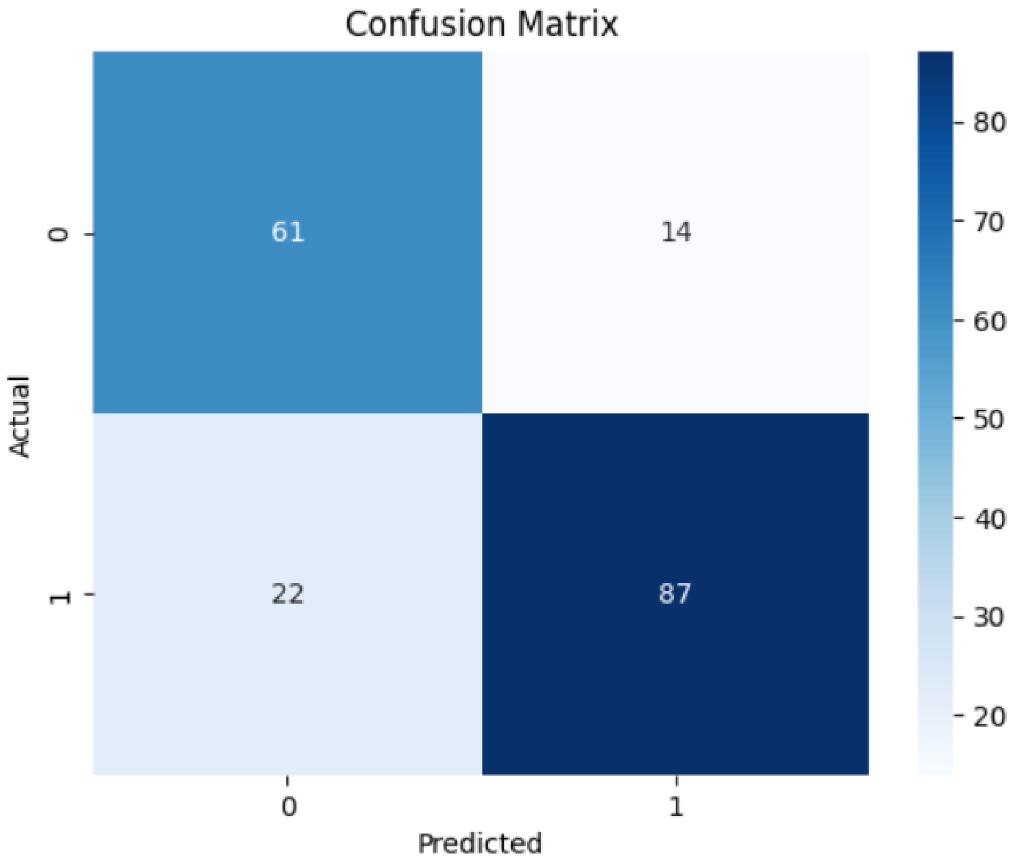
Precision

Of patients predicted to have heart disease, 86.14% actually had it

79.81%

Recall

The model identified 79.82% of all actual heart disease cases



82.86%

F1-Score

Balanced harmonic mean of Precision and Recall

0.900

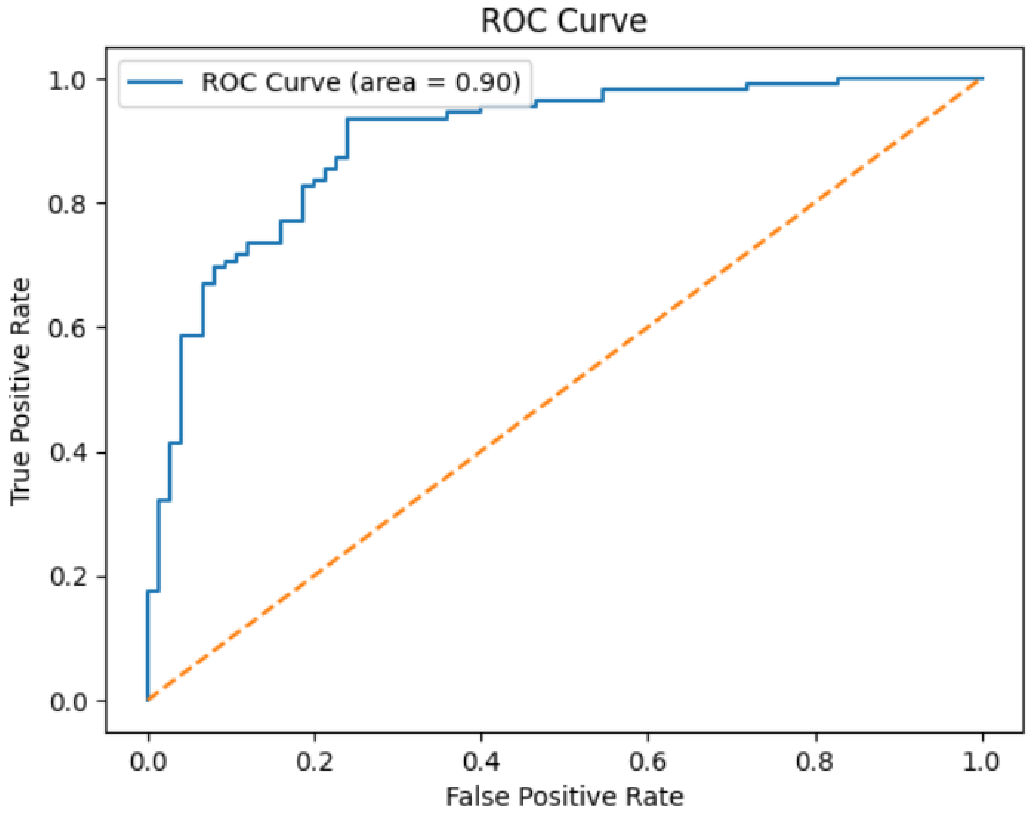
ROC-AUC

Strong discriminative ability — well above the 0.5 random baseline

19.57%

Error Rate

Misclassification rate — **1 - Accuracy**



Non-Linear Model: Random Forest

What is Random Forest?

Random Forest is a powerful **ensemble learning method** that constructs multiple decision trees during training and outputs the mode of their individual predictions. It is one of the most robust and widely-used algorithms in applied machine learning.

Each tree in the forest is trained on a **bootstrap sample** (random sample with replacement) of the training data, and at each split, only a random subset of features is considered. This dual randomness — in data and features — ensures that individual trees are **decorrelated**, and their collective vote is far more accurate and stable than any single tree.

Key Properties

- **Non-linear:** Captures complex, non-linear relationships without manual feature engineering
- **Robust to outliers:** Ensemble averaging reduces the impact of noisy data points
- **Handles mixed data:** Works well with both continuous and categorical features
- **Built-in feature importance:** Provides interpretability through feature ranking
- **Resistant to overfitting:** Ensemble approach naturally regularizes the model

Why Random Forest for Heart Disease?

Medical data is inherently complex. The relationship between clinical features (age, cholesterol, blood pressure, chest pain type) and heart disease risk is **highly non-linear** and involves intricate interactions between variables. A single decision tree might capture some of these patterns, but a Random Forest aggregates the wisdom of many trees to produce a far more accurate and reliable prediction.

Ensemble Advantage

Where Logistic Regression fits a single linear boundary, Random Forest builds **hundreds of decision boundaries** and combines them. This allows it to model the complex, multi-dimensional decision surface that separates patients with and without heart disease — something a linear model fundamentally cannot do without extensive feature engineering.

Random Forest Performance

The Random Forest model was evaluated on the same held-out test set using identical metrics. The results demonstrate **significantly stronger performance** across all five metrics compared to the Logistic Regression baseline, validating the advantage of non-linear modeling for this clinical prediction task.

86.96%

Accuracy

Overall correctness — 86.96% of test predictions were correct (+6.5% over LR)

92.08%

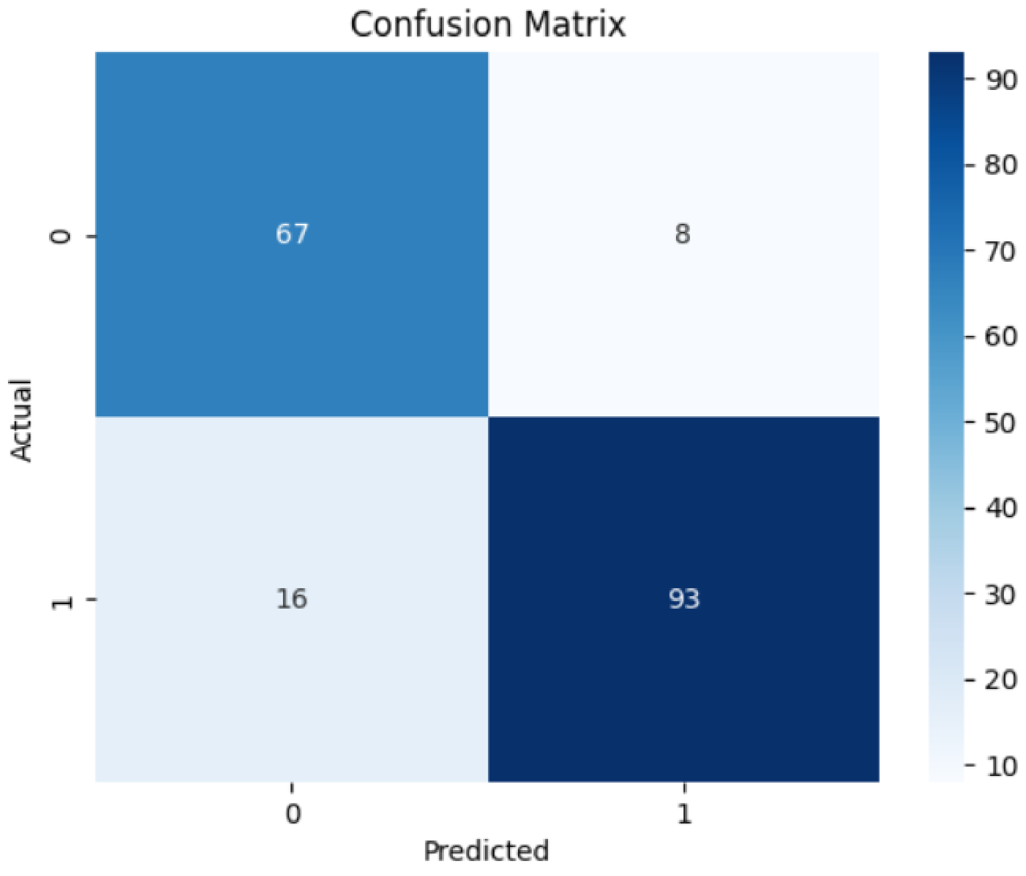
Precision

Of patients predicted to have heart disease, 92.08% actually had it (+5.9% over LR)

85.32%

Recall

The model identified 85.32% of all actual heart disease cases (+5.5% over LR)



88.57%

F1-Score

Balanced harmonic mean of Precision and Recall (+5.7% over LR)

0.921

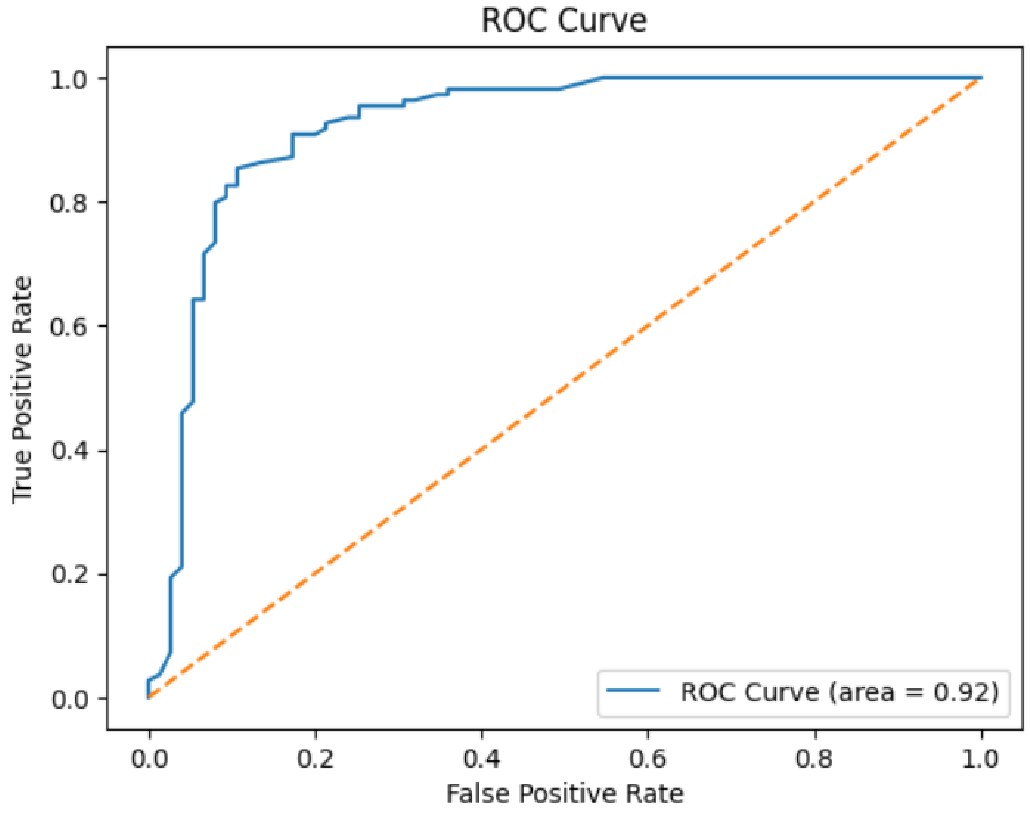
ROC-AUC

Superior discriminative ability — 0.921 vs 0.900 for Logistic Regression

13.04%

Error Rate

Misclassification rate — **1 - Accuracy**



Logistic Regression vs. Random Forest

A direct comparison of model performance on key evaluation metrics:

Metric	Logistic Regression	Random Forest
Accuracy	80.4%	87.0%
Precision	86.1%	92.1%
Recall	79.8%	85.3%
F1-Score	82.9%	88.6%
ROC-AUC	0.900	0.921

This comparison clearly demonstrates that the **non-linear Random Forest model consistently outperforms the linear Logistic Regression model** across all evaluated metrics, highlighting the benefits of capturing complex relationships in heart disease prediction.

Performance Thresholds for This Dataset

Metric	Good	Excellent
Accuracy	>80%	>85%
Precision/Recall	>85%	>90%
F1-score	>80%	>85%
ROC-AUC	>0.8	>0.9

Conclusion & Key Takeaways

Our analysis unequivocally demonstrates the **Random Forest model's superior performance** for heart disease prediction within this dataset. With an impressive **accuracy of ~87%** and an **ROC-AUC of 0.92**, it significantly outperforms the linear Logistic Regression model.

This means the Random Forest model is much better at successfully predicting heart disease while minimizing false positives and false negatives, which is crucial in clinical applications. This study underscores the fundamental principle that **evaluating multiple models is critical** in data science to guarantee the most robust and impactful outcomes.

Thank you.